ISE 3200 - HW3: Duality & Sensitivity

Priya Natarajan

Q1. Primal: max z = 4x1 + 3x2 subject to 2x1 + x2 <= 10, x1 + 3x2 <= 15, x1, x2 >= 0.

Q1a) Write the dual LP.

min w = 10y1 + 15y2 s.t. 2y1 + y2 >= 4, y1 + 3y2 >= 3, y1, y2 >= 0

Q1b) Use strong duality to give the dual optimal value.

w* = z* = 24 (primal optimum x = (3, 4))

Q2) Given x* = (3, 4), use complementary slackness to find y*.

Both x > 0, so both dual constraints are tight: y* = (9/5, 2/5).

Q3. Use the primal from Q1 and its optimal basis.

Q3a) What is the shadow price of the first resource? Interpret it.

y1* = 1.8: one more unit of resource 1 raises z by about 1.8.

Q3b) Over what range of c1 does the current basis stay optimal?

1 <= c1 <= 6